

CORRELATION VS. CAUSATION

Module 4: The Advanced Masterclass

1. THE CORRELATION TRAP

The most common mistake in data science is assuming that because two variables move in sync, one must be *causing* the other.

- **Correlation (r):** A measure of the linear relationship between two variables. It ranges from -1 to $+1$.
- **Causation:** A demonstration that a change in variable A *directly produces* a change in variable B .

The Golden Rule: Correlation is a necessary condition for causation, but it is **not** sufficient.

2. SPURIOUS CORRELATIONS

Some correlations are so tight they look like absolute proof, but they are entirely accidental.

- *Example:* The number of ice cream sales perfectly predicts the number of shark attacks.
- *The Truth:* Neither causes the other. A third, invisible variable—**Temperature**—is causing both to rise.

3. THE CONFOUNDING VARIABLE (THE HIDDEN Z-AXIS)

The Confounding Variable (or “Lurking Variable”) is the hidden factor that influences both your independent and dependent variables, creating an illusion of a relationship

where none exists.

- X (**Ice Cream**) and Y (**Shark Attacks**) seem linked.
- Z (**Temperature**) is the hidden Confounding Variable.

To break the illusion, you must “control” for Z . In data science, this means splitting your data into segments based on Z . Once you look at ice cream sales *only on hot days* and *only on cold days* separately, the correlation to shark attacks vanishes.

4. SIMPSON’S PARADOX

This is the ultimate statistical illusion. Simpson’s Paradox occurs when a trend appears in several different groups of data but disappears or reverses when these groups are combined.

- **The Takeaway:** Always check if your data was “aggregated.” If you combine data from fundamentally different populations, the aggregate trend is almost guaranteed to lie to you.

5. HOW TO PROVE CAUSALITY

If you can’t trust correlation, how do we prove anything?

1. **Randomized Controlled Trials (RCTs):** The gold standard. By randomly assigning participants to groups, you “shuffle away” the effect of any potential confounding variables.
2. **Temporal Precedence:** X must happen *before* Y .
3. **Mechanism:** You must be able to explain the physical or logical process that connects X to Y .